POSB

People's Own Savings Bank

Applicant Import API

Technical Integration Guide

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Classification: Internal / Developer Use Only

Prepared by: Digital Banking Team
System: Microsoft Dynamics 365 Finance & Operations
`Service: POSB_ApplicantImportService`

1. Overview

The POSB Applicant Import API provides a secure, structured endpoint for programmatically submitting job applicants into the Dynamics 365 Finance & Operations (D365FO) Human Capital Management (HCM) module. It is designed for integration with external recruitment platforms, portals, or batch import processes.

The API creates or identifies a person record in the Global Address Book, creates an HCM applicant profile, links a job application to a configured recruitment project, and optionally attaches a CV/resume document.

> **Key Behaviour:** If a person already exists with the same First Name, Last Name, Date of Birth, and Email, the service treats it as an existing applicant and creates a new application against that applicant — no duplicate person record is created.

1.1 Technology

- Platform: Microsoft Dynamics 365 Finance & Operations
- Service Class: POSB_ApplicantImportService
- Protocol: SOAP / OData via D365FO Custom Service
- Authentication: Azure Active Directory (OAuth 2.0 Bearer Token)
- Data Format: JSON

2. Endpoint Details

2.1 Service Information

Field Name	Type	Required	Example	Description
`Method`	POST	—	—	HTTP method
`Content-Type`	Header	**Yes**	application/json	Must be set on every request
`Authorization`	Header	**Yes**	Bearer <token>	Azure AD OAuth 2.0 token

2.2 URL Pattern

```
POST https://posb-
test.sandbox.operations.sa.dynamics.com/api/services/ApplicantImportServiceGroup/PosbApplicantImp
ortService/SubmitApplicant
```

3. Request Payload

3.1 Structure

All fields must be wrapped inside a contract object at the root of the JSON body. Field names are PascalCase and are case-sensitive.

```
{
  "contract": {
    "FirstName":    "string",
    "LastName":     "string",
```

```
    "MiddleName":   "string",
    "Email":        "string",
    "Phone":        "string",
    "BirthDateUtc": "datetime",
    "Gender":       "string",
    "Citizenship":  "string (ISO-3)",
    "MaritalStatus":"string",
    "StreetAddress":"string",
    "City":         "string",
    "ZipCode":      "string",
    "Country":      "string (ISO-2 or ISO-3)",
    "CurrentJobTitle":           "string",
    "EducationLevelDescription": "string",
    "ApplicantId":               "string",
    "RecruitingId":              "string",
    "ExternalApplicationId":     "string",
    "CoverLetter":               "string",
    "FileName":                  "string | null",
    "FileBytes":                 "string (Base64) | null"
  }
}
```

3.2 Field Reference

Field Name	Type	Required	Example	Description
`FirstName`	string	**Yes**	TENDAI	Applicant's first name
`LastName`	string	**Yes**	MOYO	Applicant's last name
`MiddleName`	string	No	BLESSING	Applicant's middle name (empty string if none)
`Email`	string	**Yes**	t.moyo@bank.com	Primary email. Used for duplicate detection
`Phone`	string	No	+263772123456	Phone number. No spaces recommended
`BirthDateUtc`	datetime	**Yes**	1988-03-22T00:00:00Z	Date of birth in ISO 8601 UTC format. Used for duplicate detection
`Gender`	string	No	Male / Female	Accepted values: Male, Female. Defaults to NonSpecific
`Citizenship`	string	No	ZWE	ISO 3-letter country code for citizenship
`MaritalStatus`	string	No	Single	Accepted: Single, Married, Divorced, Widowed
`StreetAddress`	string	No	45 Samora Machel Ave	Street address line
`City`	string	No	Harare	City or town
`ZipCode`	string	No		Postal/ZIP code (may be empty)
`Country`	string	No	ZW	ISO 2 or 3 letter country code for postal address
`CurrentJobTitle`	string	No	Software Engineer	Applicant's current job title
`EducationLevelDescription`	string	No	Degree/Diploma	Must match an existing HcmEducationLevel description in D365
`ApplicantId`	string	**Yes**	POSB-AI-00007	Applicant ID to assign. Must be unique and follow POSB numbering

Field Name	Type	Required	Example	Description
`RecruitingId`	string	No	IT Test Project	ID of an existing D365 recruitment project. Project must have Status = Started
`ExternalApplicationId`	string	No	EXT-003	Reference ID from the external system for traceability
`CoverLetter`	string	No	I am applying...	Cover letter text
`FileName`	string	No	CV_Moyo.pdf	File name including extension. Required if FileBytes is provided
`FileBytes`	string	No	JVBERi0x...	Base64-encoded file content. Supported types: PDF, DOCX, DOC, JPG, PNG

Important — ApplicantId: The ApplicantId must be provided by the calling system and must be unique. POSB's standard sequence format is POSB-AI-XXXXX (e.g. POSB-AI-00007). If the provided ID is already in use, the request will fail with a duplicate error.

Important — RecruitingId: The recruitment project referenced by RecruitingId must exist in D365 and must have its Status set to Started. Providing an ID for a project in any other status (e.g. Created, Finished) will result in an error.

Important — EducationLevelDescription: This value must exactly match a Description field in the HcmEducationLevel table in D365. If no match is found, the education level field on the applicant will be left blank — no error is thrown.

4. Sample Requests

4.1 Full Request (No Attachment)

```
{
  "contract": {
    "FirstName":     "TENDAI",
    "LastName":      "MOYO",
    "MiddleName":    "BLESSING",
    "Email":         "tendai.moyo@posbtest.com",
    "Phone":         "+263772123456",
    "BirthDateUtc": "1988-03-22T00:00:00Z",
    "Gender":        "Female",
    "Citizenship":   "ZWE",
    "MaritalStatus":"Married",
    "StreetAddress":"45 Samora Machel Avenue",
    "City":          "Harare",
    "ZipCode":       "",
    "Country":       "ZW",
    "CurrentJobTitle":            "Software Engineer",
    "EducationLevelDescription": "Degree",
    "ApplicantId":                "POSB-AI-00007",
    "RecruitingId":               "REC-2026-001",
    "ExternalApplicationId":      "EXT-002",
    "CoverLetter":   "I am excited to apply for this position at POSB.",
    "FileName":      null,
    "FileBytes":     null
  }
}
```

4.2 Full Request (With CV Attachment)

When attaching a file, encode the file content as Base64 and include it in FileBytes. The FileName must include the correct file extension.

```
{
  "contract": {
    "FirstName":    "SARAH",
    "LastName":     "DUBE",
    "MiddleName":   "",
    "Email":        "sarah.dube@posbtest.com",
    "Phone":        "+263715551234",
    "BirthDateUtc": "1995-03-20T00:00:00Z",
    "Gender":       "Female",
    "Citizenship":  "ZWE",
    "MaritalStatus":"Single",
    "StreetAddress":"456 API Test Avenue",
    "City":         "Harare",
    "ZipCode":      "",
    "Country":      "ZW",
    "CurrentJobTitle":           "Software Engineer",
    "EducationLevelDescription": "Degree",
    "ApplicantId":               "POSB-AI-00008",
    "RecruitingId":              "REC-2026-001",
    "ExternalApplicationId":     "EXT-003",
    "CoverLetter":  "I am writing to express my interest in this position.",
    "FileName":     "SarahDube_CV.pdf",
    "FileBytes":    "JVBERi0xLjQ..."
  }
}
```

Base64 Encoding: To generate a Base64 string from a file you can use: PowerShell: [Convert]::ToBase64String([IO.File]::ReadAllBytes('C:\path\to\file.pdf')) | Linux/Mac: base64 -i file.pdf

5. Response

5.1 Response Structure

All responses return HTTP 200. Inspect the parmStatus field to determine success or failure.

```
{
  "$id":                  "1",
  "parmStatus":           "Success | Duplicate | Error",
  "parmMessage":          "string",
  "parmApplicantId":      "string",
  "parmApplicantRecId":  0,
  "parmApplicationRecId": 0
}
```

5.2 Response Fields

Field	Type	Description
parmStatus	string	Outcome of the request. Values: Success, Duplicate, Error
parmMessage	string	Human-readable description of the result or error

Field	Type	Description
`parmApplicantId`	`string`	The ApplicantId assigned in D365 (e.g. POSB-AI-00007). Empty on error
`parmApplicantRecId`	`Int64`	Internal D365 RecId of the HcmApplicant record. 0 on error
`parmApplicationRecId`	`Int64`	Internal D365 RecId of the HRMApplication record. 0 on error

5.3 Status Values

Status Value	Meaning
Success	Applicant and application were created or updated successfully. All RecIds are populated.
Duplicate	A person with matching Name + Email + DOB already exists. A new application was still created against the existing applicant.
Error	Processing failed. See parmMessage for the specific reason. No records are committed (full rollback).

5.4 Sample Success Response

```
{
  "$id":                   "1",
  "parmStatus":            "Success",
  "parmMessage":           "Applicant and application processed.",
  "parmApplicantId":       "POSB-AI-00007",
  "parmApplicantRecId":    5637152826,
  "parmApplicationRecId": 5637154326
}
```

5.5 Sample Error Response

```
{
  "$id":                   "1",
  "parmStatus":            "Error",
  "parmMessage":           "Recruiting project REC-2026-001 has status Finished. Only
Started projects accept applications.",
  "parmApplicantId":       "",
  "parmApplicantRecId":    0,
  "parmApplicationRecId": 0
}
```

6. Error Reference

6.1 Common Validation Errors

Field Name	Type	Required	Example	Description
`FirstName / LastName missing`	Error	**Yes**	—	Both fields are mandatory. Empty strings are rejected
`Email missing`	Error	**Yes**	—	Required for duplicate detection and contact creation
`BirthDateUtc missing`	Error	**Yes**	—	Required for duplicate detection
`ApplicantId already in use`	Error	**Yes**	—	The provided ApplicantId exists on another HcmApplicant record

Field Name	Type	Required	Example	Description
`RecruitingId not found`	Error	No	—	No HRMRecruitingTable record matches the provided RecruitingId
`RecruitingId wrong status`	Error	No	—	Recruitment project exists but is not in Started status

6.2 Deserialization Errors

If the JSON body cannot be parsed, D365 returns an HTTP-level error (not the standard response format):

```
{
  "Message": "An exception occured when deserializing the request...",
  "ExceptionType": "XppServicesDeserializationException",
  "ActivityId": "..."
}
```

Common causes:

- Missing Content-Type: application/json header
- Field names not in PascalCase (e.g. sending firstName instead of FirstName)
- contract wrapper key missing or misspelled
- Invalid JSON syntax

7. Integration Notes & Behaviour

7.1 Duplicate Detection Logic

Before creating any records, the service checks for an existing person using a combination of three fields:

- First Name + Last Name (exact match, case-sensitive)
- Date of Birth (day, month, year)
- Email address (exact match)

If all three match an existing person who already has an HcmApplicant record, a new HRMApplication is created against that applicant. No new person or applicant record is created.

7.2 Application ID Generation

The service auto-generates a unique application ID for each HRMApplication record using the format:

```
POSB-EX-XXXXX    (e.g. POSB-EX-00001, POSB-EX-00002, ...)
```

This ID is separate from the ExternalApplicationId provided by the caller. ExternalApplicationId is stored for cross-reference with the originating system.

7.3 Transactional Safety

All database operations within a single request are wrapped in a single transaction (ttsbegin / ttscommit). If any step fails, the entire transaction is rolled back. Partial records will never be left in the database.

7.4 Education Level Matching

The EducationLevelDescription value is matched against the Description field of the HcmEducationLevel table. The match is exact and case-sensitive. If no match is found, the applicant is still created — the education level field is simply left blank. To avoid this, confirm the exact description values available in D365 with the HR team before integration.

7.5 File Attachments

Files are attached to the HRMApplication record using D365's document management framework. Supported file types include PDF, DOCX, DOC, JPG, and PNG. The file must be encoded as a Base64 string. Both FileName and FileBytes must be provided together — providing one without the other will result in no attachment being created.

> **Note:** File attachment support is currently being finalised. Once confirmed, this guide will be updated with tested examples. All other fields are fully operational.

8. Prerequisites & Setup Checklist

Before calling the API, ensure the following are in place in D365:

1. Authentication token obtained from Azure AD for the D365 environment
2. RecruitingId of an active recruitment project with Status = Started confirmed with HR
3. EducationLevelDescription values confirmed against HcmEducationLevel table in D365
4. ApplicantId numbering scheme agreed (format: POSB-AI-XXXXX) and sequence coordinated with D365 team to avoid conflicts
5. Content-Type: application/json header included in every request
6. Service endpoint URL confirmed for the target environment (UAT / Production)

9. Support & Contact

For integration queries or issues, contact:

Team	Digital Banking / D365 Development Team
System	Microsoft Dynamics 365 Finance & Operations
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Status	Draft — Pending file attachment confirmation

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